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Dietary Intake and Body Composition During Ramadan in Athletes: A Systematic Review and Meta-Analysis With Meta-Regression

Khaled Trabelsi, Achraf Ammar, Omar Boukhris, Jordan M Glenn, Cain C T Clark, Stephen R Stannard, Gary Slater, Piotr Żmijewski, Tarak Driss, Helmi Ben Saad, Karim Chamari, Hamdi Chtourou

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Research output: Contribution to journal › Review article › peer-review

9 Citations (Scopus)

- Overview
- Fingerprint

Abstract

The aim of this research was to evaluate the effect of Ramadan observance on dietary intake and body composition in adult athletes. This was a systematic review and meta-analysis. Data sources used were PubMed, Web of Science, Scopus, and Taylor and Francis. Eligibility criteria for selecting studies were as follows: single-group, pre-/post-Ramadan, with or without control group, conducted in athletes aged ≥19 years training at least twice a week, and published in any language before August 25, 2021. Studies assessing dietary intake and/or body composition were deemed eligible. The methodological quality of studies was assessed using QualSyst. Nine studies evaluated dietary intake; 4 of these were rated as strong quality and the remaining as moderate. Of the 17 selected studies evaluating body composition, 7 were of strong quality and the remaining 10 were rated as moderate. Compared to pre-Ramadan, energy (number of studies, = 7; number of participants, N = 78; = -0.781; 95% confidence interval [CI], -1.416 to -0.145; = 0.016), carbohydrate (= 5; N = 50; = -1.643; 95% CI, -2.949 to -0.336; = 0.014), and water (= 4; N = 39; = -1.081; 95% CI, -1.371 to -0.790; = 0.000) intakes decreased during Ramadan. However, fat (= 5; N = 50; = -0.472; 95% CI, -1.085 to 0.140; = 0.131) and protein (= 5; N = 60; = -0.574; 95% CI, -1.213 to -0.066; = 0.079) intakes remained unchanged. Compared to pre-Ramadan, body mass (= 16; N = 131; = -0.262; 95% CI, -0.427 to -0.097; = 0.002) and body fat percentage (= 8; N = 81; = -0.197; 95% CI, -0.355 to -0.040; = 0.014) decreased in the fourth week of Ramadan. Lean body mass did not change during Ramadan (= 4; N = 45; = -0.047; 95% CI, -0.257 to 0.162; = 0.658). Carbohydrate and total water intake decreased with the observance of Ramadan, but fat and protein intake were unchanged. Continued training of athletes during Ramadan was associated with a decreased body mass and body fat percentage, but not lean body mass, toward the end of the fasting month. Key pointsRamadan fasting decreases body mass and body fat percentage of athletes, but not lean body mass.Longer durations of fasting could provoke more pronounced decrements in body mass.Carbohydrate and total water intake decreased during Ramadan in athletes.Future studies, with greater methodological rigor, are required to better discern changes in dietary intake and body composition during Ramadan. Supplemental data for this article is available online at https://doi.org/10.1080/07315724.2021.2000902 .

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- Athletes
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- lean mass
- macronutrients

ASJC Scopus subject areas

- Medicine (miscellaneous)
- Nutrition and Dietetics

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Other files and links

- Link to publication in Scopus

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author = "Khaled Trabelsi and Achraf Ammar and Omar Boukhris and Glenn, {Jordan M} and Clark, {Cain C T} and Stannard, {Stephen R} and Gary Slater and Piotr {\.Z}mijewski and Tarak Driss and {Ben Saad}, Helmi and Karim Chamari and Hamdi Chtourou",
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T1 - Dietary Intake and Body Composition During Ramadan in Athletes

T2 - A Systematic Review and Meta-Analysis With Meta-Regression

AU - Trabelsi, Khaled

AU - Ammar, Achraf

AU - Boukhris, Omar

AU - Glenn, Jordan M

AU - Clark, Cain C T

AU - Stannard, Stephen R

AU - Slater, Gary

AU - Żmijewski, Piotr

AU - Driss, Tarak

AU - Ben Saad, Helmi

AU - Chamari, Karim

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KW - Athletes

KW - adults

KW - energy intake

KW - fat mass

KW - intermittent fasting

KW - lean mass

KW - macronutrients

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